



DiveXplore at the Video Browser Showdown 2025

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Abstract. The diveXplore system has been participating in the VBS and IVR4B competitions for the last few years and achieved decent performance. However, due to our experience with novices at VBS2024 and IVR4B2024, we realize that the system can be significantly improved in terms of usability, which is our main focus this year. Therefore, we integrate a direct video scrubbing feature (via mouse movement in the keyframe area), provide better context-visualization of keyframes, implement collaboration features for cooperative use in a team, allow textual search inside a single video, and conduct a qualitative evaluation of text query performance with CLIP, which gives insights into dos and don'ts.

Keywords: video retrieval · interactive video search · video analysis

1 Introduction

The diveXplore system has been a competitor at the Video Browser Showdown (VBS) for several years [2, 6, 7] and has undergone various iterations in its UI and feature set. A separate version of diveXplore, diveXB, was created to compete at the IVR4B2023 [5] and IVR4B2024 [3] evaluations, which are particularly challenging since they focus on novice users only. This system aimed to create a separate version of diveXplore, offering a more intuitive and easy-to-use interface. In both competitions, as well as at the VBS2024 [6], our system was very competitive and achieved great results (*Best QA/Novices* at VBS2024, *Best IVR4B2023* system, *3rd best IVR4B2024* system). For this year, we combine the two different versions (diveXplore and diveXB) to benefit from their separate strengths and to simplify development, and integrate feedback from both novices and experts. Our recent modifications include various design changes and new features to ease the collaboration between users in the same team. Additionally, after inspecting the text queries used by novices and experts at IVR4B2024 and VBS2024 and experimenting with CLIP models, we recognize a significant potential in optimizing query strings. Therefore, we perform an evaluation of different types of text queries (keywords vs. full descriptions) for a subset of tasks performed at VBS and IVR4B and draw conclusions for the use of our system.

2 VBS2025

In this year’s VBS, three separate datasets will be used to evaluate the systems. The datasets used and the number of files and video length can be seen in Table 1. In contrast to last year’s competition, the third shard of the V3C dataset (V3C3) will also be used, adding another 11,215 videos. As in previous years, the competition will include different query types to evaluate the performance of the competing systems.

Table 1. Datasets used for VBS2025 with corresponding total files and hours of video.

Dataset	Total Files	Hours of Video
V3C1+V3C2+V3C3 [4]	28450	3800
Marine Video (Underwater) [8]	1374	12.36
LapGynLHE (Laparoscopic Gynecology)	78	104.77

The first tested query type at VBS2025 is *known item search* (KIS), where competitors must retrieve keyframes from specific video segments, divided into visual and textual queries. Visual KIS tasks require the user to submit a keyframe present in a shown video segment. Textual KIS tasks involve the retrieval of some information based on a question or statement given in the task description.

Additionally, *ad-hoc search tasks* (AVS) and *question-answering* (QA) tasks are tested. AVS tasks involve finding as many keyframes as possible matching a textual description, while QA queries require submitting a textual answer based on a question related to content in one or several videos.

Reward points are weighted by submission speed or the number of correct videos (for AVS tasks), so improving system usability and performance is the highest priority. As diveXplore already received new features to compete at IVR4B, we focused on refining existing features, as well as optimizing the handling of these features (like query optimization for CLIP) to enhance our system’s competitiveness.

The main contributions for this iteration are (i) the addition of collaboration features between users of the same system, (ii) the integration of feature requests based on feedback from IVR4B, and (iii) an analysis of query strings and corresponding results, to learn how users optimally employ the CLIP-based search in our system.

3 DiveXplore at the VBS2025

Following the IVR4B2024 competition, diveXplore has undergone significant changes in its feature set and user interface [3]. Most significant are the changes to the context-visualization of keyframes when selected by the user, as well as the addition of a video scrubbing feature. To improve the context-visualization,

we included several new preview windows showing video summaries, the video itself, and visually similar videos. The scrubbing feature allows a user to quickly scrub through a video segment by dragging the mouse across the vertical axis of a keyframe while pressing a specified key. The implementation of this feature, as well as the reworked context-visualization, can be seen in this year’s diveXplore paper for the IVR4B. [3] These features proved to be effective, as diveXplore was able to compete directly with the competition’s best systems in a very close race. To further enhance the efficiency and effectiveness of our system, we decided to refine present features based on user feedback, incorporate new features aiding the collaborative nature of the VBS, and refine the way we use our system.

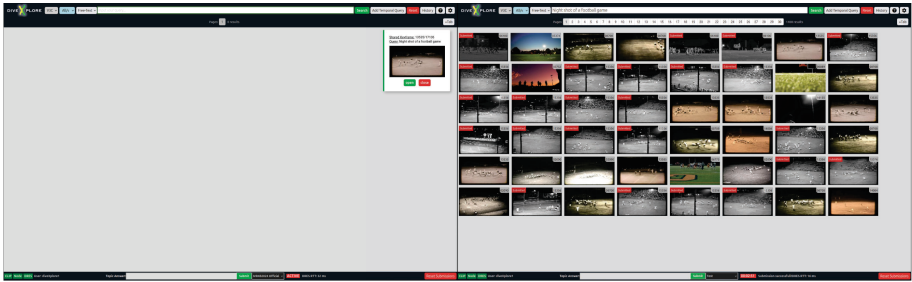


Fig. 1. Collaboration features consists of a *keyframe-share* functionality (left) and a *shared submission view* (right). The keyframe-share functionality consists of a popup window opening after another user shared a keyframe (left). The receiving user can see the keyframe ID and the query to get to it. After clicking on open, a new window displays the shotlist of the chosen keyframes’ video. In the shared submission view, all users see already submitted keyframes to avoid duplicate submissions in the team.

3.1 Collaboration Features

At the VBS competition, multiple users can compete for each system. With previous versions of diveXplore, two users were operating the system at once on separate PCs. Due to the vast amount of different approaches a user might take to fulfill a task, we deemed it necessary to implement a communication method between the users that does not distract from the main usage of the system. To achieve this, we implement two new collaboration features in diveXplore.

First, already submitted keyframes are marked for all users. This is done by overlaying the submitted keyframes with a grey filter and red text, signifying that this keyframe has already been handed in by a team member and can be ignored. Additionally, an opt-in setting provides the ability to completely hide these keyframes, allowing for a distraction-free retrieval experience when for example multiple keyframes have to be submitted per video.

The second collaboration feature allows the users to share found keyframes via a new button in the video preview or by clicking on the timestamp shown

on the keyframe. When a keyframe is shared, a pop-up opens on all other user’s screens showing the keyframe name, the image itself, as well as a *close* and *open* button (as shown in Fig. 1). Upon pressing the *open* button, a new tab opens showing all of the keyframes of the video the chosen keyframe is part of.

The combination of these collaboration features allows the users to seamlessly work together without interfering with the core user interface of diveXplore. By providing easy ways of sharing keyframes or notifying users of already handed-in keyframes, teamwork among participants should significantly improve.

3.2 Incorporating Feedback from IVR4B

After competing at IVR4B2024 we gathered valuable feedback from the novices operating our system. By incorporating that feedback we further improve the usability and intuitivity of our system. These changes include a rework of window sizes for the video preview (to avoid the window header shifting on changing tabs), the inclusion of our video scrubbing feature on the shotlist page, as well as various minor design changes.

One major change is the ability to perform textual queries within a single video. Due to the size of the shotlist on some long videos from the V3C dataset, novices were looking for a way to query within such a shotlist to find specific keyframes faster. Now, when entering the shotlist page of a video, a separate query area allows the user to perform CLIP queries that are restricted to the current video’s content. Additionally, the video scrubbing feature introduced for the IVR4B competition [3] has been integrated into the shotlist view, allowing users to quickly grasp the context in a specific video segment by hovering the mouse over a keyframe. Allowing the user to quickly retrieve information about a video’s context proved to be crucial during the IVR4B. Because of this, we also expand the number of keyframes retrieved immediately when opening the shotlist preview and exploration preview in the video preview windows.

4 Analyzing CLIP Query Performance

As some of the tasks used in the IVR4B competition were the same as in VBS2024, we compare the performance of queries used by the team members in



Fig. 2. Keyframes inside the target video segments for three tasks from VBS2024 and IVR4B2024. The video IDs are (from left to right): 03757, 13874, and 07601.

Table 2. Query strings and the associated rank of the target keyframe per user. *diveXplore1* and *diveXplore2* are experts who competed in the VBS2024, user *Novice* is a novice who competed in the IVR4B2024. All query strings were taken from the competition logs and grammatical/spelling errors are consistent with the queries used.

Task	User	Query Text	Rank
T1	Optimal query	Black and white mural with a ladder	1
	diveXplore1	grayscale video with a person on a ladder and some creative wall paointing	9
	Novice	person on ladder grey scale painting person on ladder grey scale grafitti	11 -
	diveXplore2	a man painting the room with two windows a black and white	-
T2	Optimal query	Selfie angle of a man in blue shirt hiking steep rocky grassy mountain	1
	diveXplore1	hiker in blue t-shirt and red bag at mountain summit in a sunny weather	15
		hiker with sunglasses and blue t-shirt and red bag at mountain summit in a sunny weather	16
	Novice	man blue t-shirt sunglasses mountain outdoor	23
	diveXplore2	a hiker reach the summit < the hikers that carry their backpacks	-
T3	Optimal query	An aerial view of white cliffs along the coast with green fields in the background	1
	diveXplore2	white cliffs in front of sea	8
	Novice	cliff sea flying golf	-
	diveXplore1	-	-

each competition to an “optimal query”. These optimal queries were initially generated using ChatGPT¹ to analyze the images and provide efficient descriptions. They were then refined through iterative adjustments by adding, removing, or changing specific terms, to achieve better rankings (position of the keyframe in the query results) of the target keyframe. Using ChatGPT to generate the initial prompt proved very efficient, as the queries it returned achieved higher rankings than our predefined queries in all instances.

Users sometimes retrieved the correct keyframe as the first result, even when their query was relatively vague, likely due to chance. To address this, we defined the “optimal query” based on how many keyframes from the correct video appeared on the first page of the results (42 results per page), rather than just the first position. The complete set of queries for three of the analyzed tasks can be seen in Table 2. We use OpenCLIP ViT-H/14, trained on the LAION2B dataset [1], for our CLIP queries.

¹ <https://openai.com/chatgpt>.

The results for Task 1 (see Table 2 and video ID 03757 in Fig. 2) show a significant impact on the ranking of the correct keyframe based on the wording and semantics of the sentence. *diveXplore2*'s query is semantically incorrect, as it suggests, that a man painted the room utilizing two windows, which is also reflected in the absence of the keyframe in the first 42 results. The terms "painting" and "graffiti" also negatively impact the query result as they are too vague or connected to different scenarios like street art. *diveXplore1* achieved the best results but contained a spelling error ("paointing" instead of "painting"), which when corrected improves the ranking by three places. Filler words like "some" also hurt the scoring, as excluding this single word leads to the keyframe placing on the fourth rank.

Task 2 (see Table 2 and video ID 13874 in Fig. 2) shows the importance of semantics again. *diveXplore1*'s query mentions a "hiker in blue t-shirt and a red bag", which could be interpreted as a hiker in a red bag or a separate red bag in the frame. Excluding "red bag" from the query string improves the rating by placing the keyframe on the fifth rank. The novice query string did not perform well, as it has no semantics and does not mention the main activity in the keyframe, which is hiking (as seen in Fig. 2). Adding semantics and the term "hiking" improves the rating to the tenth rank. (Query used: "man with blue t-shirt wearing sunglasses on a mountain hiking")

For Task 3 (see Table 2 and video ID 07601 in Fig. 2), only *diveXplore2* managed to retrieve the correct keyframe in the first 42 results. The novice user added additional information which is temporally spaced apart from the main keyframe, leading to retrieved keyframes including golf scenes, as well as cliff and sea scenes. *diveXplore2*'s query was accurate but too vague since the dataset contains many different scenes of cliffs next to the sea. The optimal query further shows the significance of including terms for the camera positioning in an image (mentioning "aerial view"), which is also supported by the optimal query for Task 2.

Across all tasks analyzed, we noticed a significant improvement in retrieval performance when full, semantically correct sentences were used in a query string, instead of using keyframes that are not semantically connected. Adding information about the camera view, as well as avoiding the usage of filler words is essential for a good retrieval performance. Furthermore, it seems crucial to restrict the query to the most significant objects/areas of the targeted keyframe instead of including minor details, as this might lead to the retrieval of keyframes where these parts have a more substantial presence on the keyframe. Based on these findings, we plan to implement a dictionary check that will analyze the query string in real time, highlighting filler words or irrelevant details and suggesting replacements with more significant terms. This should help users to optimize their query strings to improve overall retrieval performance.

5 Conclusion

We introduce *diveXplore* as our competitor system at VBS 2025. This year's iteration is built upon the feedback gathered from the IVR4B and includes new

features to ease the collaboration efforts between users at the competition. As the system proved to be very competitive at the IVR4B, the focus lied on analyzing the way the users interact with the system, mainly by analyzing the query structures most efficient in retrieving specified keyframes via CLIP.

The combination of improving the system as well as enhancing the way users interact with it will prove its effectiveness at the VBS 2025.

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